

Artificial Intelligence and Retrieval-Augmented Generation (RAG)

1. Introduction to RAG

Retrieval-Augmented Generation (RAG) is an artificial intelligence framework that combines

2. Benefits of RAG

- Reduced Hallucination: Grounding answers in retrieved factual documents drastically lowers hallucinations
- Cost Efficiency: Updating domain knowledge requires simply updating the document collection
- Source Attribution: Users can inspect the exact source chunks and pages from which the answer is generated
- Data Privacy: Proprietary or sensitive documents can be kept in a private local database

3. The RAG Pipeline Steps

Step 1: Document Ingestion - Extracting clean textual content from sources like PDF documents

Step 2: Text Chunking - Splitting long documents into manageable, overlapping passages

Step 3: Embedding and Vector Storage - Converting chunks into dense vector representations

Step 4: User Query Interface - Accepting questions via Command Line Interface (CLI) or web interface

Step 5: Context Retrieval - Computing query embeddings and retrieving the top-k most relevant chunks

Step 6: Contextual Generation - Synthesizing a grounded, accurate response using an LLM or other generative model